

Microservices Architecture Overview

This documentation provides a high-level overview of each microservice in the system. Detailed API endpoints and controller-level documentation will follow once specific controllers are provided.

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1. API Gateway

Location: /api-gateway

Responsibilities:

- Single entry point for all client requests.
- Request routing to downstream services.
- Authentication & authorization enforcement.
- Cross-cutting concerns: logging, rate limiting, retry policies, and circuit breakers.
- Aggregation of responses for composite calls.

2. Auth Service

Location: /auth-service

Base Path: /auth

Responsibilities:

- User registration and profile management for companies, universities, and students.
- Login, logout, and JWT token handling.
- Deletion of user entities (companies, universities, students).

Endpoints

Method	Path	Headers	Request Body	Response	Description
POST	/company/registration	X-User-Role: UserRoleAuthorization : Bearer <token>	RegisterRequest	UserDto	Register a new company user.
DELETE	/company/delete/{id}	X-User-Role: UserRoleAuthorization : Bearer <token>	—	HTTP 200 OK	Delete a company by its ID.
POST	/university/registration	X-User-Role: UserRoleAuthorization : Bearer <token>	RegisterRequest	UserDto	Register a new university user.

DELETE	/university/delete/{id}	X-User-Role: UserRoleAuthorization : Bearer <token>	—	HTTP 200 OK	Delete a university by its ID.
POST	/student/registration	X-User-Role: UserRoleX-User-Id: LongAuthorization: Bearer <token>	RegisterRequest	UserDto	Register a new student user.
DELETE	/student/delete/{userId}	X-User-Role: UserRoleAuthorization : Bearer <token>	—	HTTP 200 OK	Delete a student by their user ID.
POST	/login	—	AuthRequest	AuthResponse	Authenticate user and return token.

3. Company Service

Location: /company-service

Base Path: /company

Responsibilities:

- CRUD operations on company profiles and metadata.
- Searching and filtering companies by name, type, location, and industry.
- Managing favourite students for each company.
- Enforcing role-based access control (ADMIN vs COMPANY).

Endpoints: CompanyController

Method	Path	Headers	Query/Body Parameters	Response	Description
GET	/search	—	name, type, location, industry, page, size	Page<CompanyDto>	Retrieve paginated list of companies filtered by search criteria.
GET	/ownerId	—	ownerId (path)	CompanyDto	Get company details by owner's user ID.
POST	/create	X-User-Role: : UserRole	CompanyRequest (body)	CompanyDto	Create a new company profile (ADMIN only).

PUT	/update/{ownerId}	X-User-Role: UserRoleX-User-Id: Long	ownerId (path), CompanyRequest (body)	HTTP 200 OK (no body)	Update company profile by owner's ID (COMPANY only).
DELETE	/delete/{userId}	X-User-Role: UserRole	userId (path)	HTTP 200 OK (no body)	Delete company profile by owner's ID (ADMIN only).

Endpoints: FavouriteStudentController

Method	Path	Headers	Query Parameters	Response	Description
POST	/favouriteStudent/{id}	X-User-Role: UserRoleX-User-Id: Long	studentOwnerId	HTTP 200 OK (no body)	Add a student to the company's favourites (COMPANY only).
DELETE	/favouriteStudent/{id}	X-User-Role: UserRoleX-User-Id: Long	studentOwnerId	HTTP 200 OK (no body)	Remove a student from the company's favourites (COMPANY only).
GET	/favouriteStudent/{id}	X-User-Role: UserRoleX-User-Id: Long	–	List<Long>	List IDs of students favored by the company (COMPANY only).

4. Discovery Service

Location: /discovery-service

Responsibilities:

- Service registry (e.g., Eureka, Consul).
- Health checks and heartbeat monitoring of registered instances.
- Dynamic discovery of service endpoints for client services.

5. Notification Service

Location: /notification-service

Responsibilities:

- Sending notifications via email, SMS, or push channels.
- Template management and rendering for different notification types.
- Scheduling and retry logic for undelivered notifications.

6. Review Service

Location: /review-service

Base Path: /review

Responsibilities:

- Management of user reviews and ratings.
- Retrieval of paginated reviews by recipient.
- Calculation and retrieval of average ratings.
- Creation, update, and deletion of reviews.

Endpoints: ReviewController

Method	Path	Headers	Query/Path Parameters	Request Body	Response	Description
GET	/getAll/{recipientId}	–	recipientId (path) page, size (query, optional)	–	Page<ReviewDto>	Retrieve paginated list of reviews for a recipient.
GET	/getAverageRating/{recipientId}	–	recipientId (path)	–	AverageRatingResponse	Get average rating for a recipient.
POST	/add	X-User-Role: UserRole X-User-Id: Long	–	ReviewRequest	ReviewDto	Add a new review (requires appropriate user role).
PUT	/update/{id}	X-User-Role: UserRole X-User-Id: Long	id (path)	ReviewUpdateRequest	HTTP 200 OK (no body)	Update an existing review by its ID.
DELETE	/delete/{id}	X-User-Role: UserRole	id (path)	–	HTTP 200 OK (no body)	Delete a review by its ID.

		eX-User-Id: Long				
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7. Student Service

Location: /student-service

Base Path: /student

Responsibilities:

- CRUD operations on student profiles.
- Searching and filtering students by personal and academic attributes.
- Batch retrieval of student profiles.
- Enrollment and academic record management.
- Integration with Auth Service for role-based access.

Endpoints: StudentController

Method	Path	Headers	Query/Body Parameters	Request Body	Response	Description
GET	/search	–	firstName, lastName, degree, currentYear, universityId, minGpa, maxGpa, page, size	–	Page<StudentDto>	Search students by filters with pagination.
GET	/getBatch	–	page, size, studentOwnerIds (in body)	List<Long>	Page<StudentDto>	Retrieve a batch of student profiles by owner IDs.
GET	/ownerId	–	ownerId (path)	–	StudentDto	Get student details by owner's user ID.
GET	/getByUniversityId/{id}	–	id (path), page, size (query)	–	List<StudentDto>	List students enrolled in a university.
POST	/	X-User-Role: UserRole	–	StudentRequest	StudentDto	Create a new student profile (UNIVERSITY only).
DELETE	/id	X-User-Role: UserRole	id (path)	–	HTTP 200 OK (no body)	Delete student profile by owner's ID

		X-User-Id : Long				(UNIVERSITY only).
PUT	/ownerId}	X-User-Role: UserRole X-User-Id : Long	ownerId (path)	StudentRequest	HTTP 200 OK (no body)	Update student profile by owner's ID (STUDENT only).

8. University Service

Location: /university-service

Base Path: /university

Responsibilities:

- CRUD operations on university profiles and metadata.
- Management of courses, departments, and campus information.
- Association of universities with students and companies.
- Search and filtering of universities by name, type, and location.

Endpoints: UniversityController

Method	Path	Headers	Query/Path Parameters	Request Body	Response	Description
GET	/search	–	name, type, location, page, size	–	Page<UniversityDto>	Search universities with optional filters and pagination.
GET	/ownerId}	–	ownerId (path)	–	UniversityDto	Get university details by owner's user ID.
POST	/create	X-User-Role: UserRole	–	UniversityRequest	UniversityDto (201)	Create a new university profile (ADMIN only).
PUT	/update/{ownerId}	X-User-Role: UserRole X-User-Id: Long	ownerId (path)	UniversityRequest	HTTP 200 OK (no body)	Update university profile by owner's ID (UNIVERSITY only).

DELETE	/ {userId}	X-User-Role: UserRole	userId (path)	-	HTTP 200 OK (no body)	Delete university profile by owner's ID (ADMIN only).
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